Phase-Modulated Compound Sparsity: Strategy-Dependent Severity Gradients in Mathematical Reasoning Models

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Abstract

Compound sparsity—simultaneously pruning attention heads and evicting KV cache entries—is essential for efficient LLM inference but produces super-additive quality degradation whose structure is poorly understood. We show that compound severity is not phase-invariant: it is an emergent phase-strategy interaction with amplification ratios (G) from 7.5× (s=0.15, 95% CI $[4.1, 11.4]$) to 79.8× (s=0.30, 95% CI $[63.3, 96.5]$) across reasoning phases in mathematical reasoning models.

On DeepSeek-R1-Distill-Qwen-7B (500 MATH-500 problems, n_{eff}=496, $s \in \{0.15, 0.20, 0.25, 0.30\}$), importance-based eviction produces a monotonically increasing severity gradient from the high-sparsity to low-sparsity reasoning phase (median Δ: $0.39 \rightarrow 8.48$; all Bonferroni-corrected $p \ll 10^{-10}$). Random eviction develops a detectable phase gradient only above $s \approx 0.20$ (non-significant at $s \leq 0.20$; where significant, 3–39× weaker), confirming importance-based eviction as the dominant driver. A crossover emerges at s=0.25: importance-based eviction outperforms random in high-sparsity phases yet underperforms in low-sparsity phases—a qualitative reversal in strategy ranking. The pattern replicates on Qwen2.5-7B-Instruct (n_{eff}=475, Δ: $0.07 \rightarrow 3.16$; all $p < 10^{-14}$), with preliminary cross-architecture evidence from a LlamaForCausalLM model suggesting the phenomenon may extend beyond Qwen2-family architectures. We present discrimination collapse—head pruning flattening cumulative importance scores—as a correlational diagnostic: both strategies exhibit near-identical Gini drops, yet severity diverges by 15–17×, implicating the structural specificity of importance-based pruning as the key differentiator. These findings establish the phase-dependent nature of compound severity as a diagnostic foundation for future phase-aware mitigation strategies.

1 Introduction

Long-context language models place acute pressure on the key–value (KV) cache, which grows linearly with sequence length (Kwon et al., 2023; Pope et al., 2023). Two complementary responses—*attention head pruning* (Voita et al., 2019; Michel et al., 2019) and *KV cache eviction* (Li et al., 2024; Feng et al., 2025)—can be deployed simultaneously as *compound sparsity*, but the resulting quality degradation is often super-additive (Liao et al., 2026; Mozaffari et al., 2025). Recent evaluations confirm that reasoning capabilities degrade disproportionately under compound compression (von Rad et al., 2026), and that per-dimension safe zones intersect to form a smaller joint safe region (Ma et al., 2026).

Prior work treats compound interaction magnitude as roughly constant across generation (Liao et al., 2026; von Rad et al., 2026), despite evidence that pruning sensitivity depends on representational phase (Shi et al., 2026) and that no single sparse attention strategy dominates across generation stages (Nawrot et al., 2025). This assumption is problematic for reasoning models, whose generations alternate between broadly distributed attention (exploratory reasoning) and sharply concentrated attention (structured computation) (Ramachandran et al., 2026). Phase-aware methods like ThinKV (Ramachandran et al., 2026) apply different KV strategies per phase but without considering compound interaction, while importance-based eviction itself faces challenges from greedy bias (Sood et al., 2026), deferred importance (Liu et al., 2026a), and temporal dynamics in long generations (Zeng et al., 2026). These methods each optimize a single pruning axis; we characterize the interaction when multiple mechanisms are composed. If compound severity varies across these phases, a uniform sparsity budget may be safe in one phase yet destructive in another.

Why can't existing single-axis findings predict our results? HIES (Choi et al., 2025) characterizes score collapse on a single compression axis; ThinKV (Ramachandran et al., 2026) introduces phase-aware eviction without compound interaction; Nawrot et al. (2025) document strategy non-dominance across generation stages. A linear combination of these insights would predict that compound degradation either (i) uniformly amplifies, or (ii) follows the dominant single-axis pattern. Instead, we observe a *severity gradient* spanning over an order of magnitude (G=7.5–79.8$\times$ for importance vs. $G \approx 1$ for random), an effect absent from any single-axis analysis. In DeepSeek, this further manifests as a strategy ranking crossover at s=0.25; Qwen shows consistent importance dominance without crossover, indicating the crossover is model-dependent. The compound interaction exceeds the multiplicative null by 2.7$\times$ in the low-sparsity phase while falling below it in the high-sparsity phase (Appendix B). This emergent coupling is not deducible from single-axis characterizations.

We show that compound sparsity severity is strongly modulated by reasoning phase and eviction strategy. Under importance-based KV cache eviction, the Interaction Index—a measure of super-additive degradation—concentrates in low-sparsity phases and is largely absent in high-sparsity phases. On DeepSeek-R1-Distill-Qwen-7B (500 MATH-500 problems), the median phase difference rises from 0.39 at s=0.15 to 8.48 at s=0.30 (all Bonferroni-corrected $p \ll 10^{-10}$; Jonckheere-Terpstra z=33.7, $p \ll 10^{-10}$). The pattern replicates on Qwen2.5-7B-Instruct (Δ: 0.07 $\rightarrow$ 3.16; all $p < 10^{-14}$); preliminary evidence from DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM) suggests the phenomenon may extend beyond Qwen2 but at $\sim 1/40$ the magnitude and with substantially weaker strategy contrast. Random eviction develops a detectable but weak phase gradient only above $s \approx 0.20$ (3–39$\times$ smaller than importance-based), confirming that the dominant gradient is an emergent phase-strategy interaction.

We present preliminary correlational evidence for *discrimination collapse*—head pruning flattening cumulative importance scores—as a contributing factor, with the *structural specificity* of importance-based pruning (which selectively removes high-score heads) as a key differentiator from random pruning.

This work is a **diagnostic study**: we characterize a previously unknown interaction effect rather than proposing a mitigation method. Following the tradition of empirical analysis papers in NLP (Voita et al., 2019; Michel et al., 2019), our contribution is establishing and quantifying the phenomenon of phase-modulated compound severity, providing measurement tools and diagnostic findings that inform future method design. Our contributions:

1. We identify a **phase–strategy interaction**: importance-based eviction exhibits a severity gradient across reasoning phases that is 3–39$\times$ larger than under random eviction (§5).
2. We show this gradient **increases monotonically with sparsity** and replicates across two Qwen2-architecture models (DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct), with exploratory evidence from a LlamaForCausalLM model suggesting the phenomenon may extend beyond Qwen2 (§5).
3. We provide a **diagnostic analysis** revealing that aggregate score collapse (Gini) is necessary but not sufficient for severity amplification, pointing to structural specificity of pruned heads as the key differentiator (§6).

2 Related Work

Compound compression. Combining multiple compression axes produces non-additive errors. (Liao et al., 2026) report super-additive degradation when combining channel pruning with token eviction; (Mozaffari et al., 2025) formalize compound error for quantization and sparsity. (von Rad et al., 2026) evaluate compound compression across 40+ benchmarks, revealing that reasoning capabilities degrade disproportionately. (Ma et al., 2026) characterize independent phase transition points per compression dimension, implying the joint safe region is the intersection of per-dimension safe zones. GRASPrune (Wang et al., 2026) learns global budget-constrained gates for joint pruning but treats decisions as static across generation. PM-KVQ (Liu et al., 2026b) demonstrates that progressive quantization—gradually reducing bit-width over the generation process—mitigates cumulative error in long-CoT reasoning, implicitly introducing a temporal dimension to compression severity that resonates with our finding that compound interaction is phase-dependent. These works establish compound interaction but share an implicit assumption: interaction magnitude is constant across the

generation process.

KV cache eviction. Most methods rely on importance scoring whose reliability is increasingly questioned. Ada-KV (Feng et al., 2025) introduces per-head adaptive budgets; SnapKV (Li et al., 2024) compresses from a fixed observation window; (Feng et al., 2026) show worst-case loss 2.3–4.3× higher than expected under standard aggregation. (Chen et al., 2024) combine proxy-token and random eviction to counter attention bias. RetentiveKV (Liu et al., 2026a) identifies deferred importance, and (Sood et al., 2026) show that single importance metrics exhibit systematic greedy bias. (Choi et al., 2025) independently show that importance-based head discrimination collapses under heavy single-axis pruning. Our work extends this to the compound setting, revealing that collapse is amplified by the interaction of two compression axes and modulated by reasoning phase—phenomena absent in single-axis analysis. The crossover effect (§5.3) is a prediction that single-axis frameworks cannot generate. RLKV (Du et al., 2025) uses reinforcement learning to identify reasoning-critical heads and selectively preserve their KV caches, demonstrating that head-level importance varies significantly for reasoning tasks. LongFlow (Su et al., 2026) targets the long-output setting of reasoning models with an efficient importance metric derived from attention intermediates. Lethe (Zeng et al., 2026) introduces temporal adaptivity with layer-wise multi-round pruning, and SparK (Liao et al., 2026) extends eviction to the channel dimension with recoverable sparsity. MUSTAFAR (Joo et al., 2025) applies multi-stage adaptive compression for reasoning, while LKV (Anonymous, 2026) learns end-to-end head-wise budgets for KV cache allocation. Concurrent work examines KV cache compression's impact on reasoning capabilities (Liu et al., 2025), evaluating single-axis compression strategies on mathematical reasoning benchmarks; our work extends this analysis to the compound setting where head pruning and KV eviction interact. These methods each optimize a single pruning axis—whether aggregation, metric, temporal, or channel—without characterizing compound interactions when multiple mechanisms are simultaneously active.

Phase-aware inference. ThinKV (Ramachandran et al., 2026) applies different KV strategies per phase but without compound interaction. (Shi et al., 2026) find pruning sensitivity depends on "silent" vs. "decisive" representation phases. (Monjur et al., 2026) show pruning combined with test-time scaling can exceed unpruned baselines; (Nawrot et al., 2025) find no universally optimal sparse attention strategy across generation stages. (Ananthanarayanan et al., 2026) identify a sharp phase transition in single-axis KV cache compression at high sparsity, characterizing the single-dimension failure mode that our compound analysis extends across multiple axes. R-KV (Cai et al., 2025a) exploits reasoning-specific redundancy for aggressive cache compression, targeting per-axis optimization rather than characterizing compound degradation. These works establish phase-dependent compression tolerance along single axes. Our work bridges this by quantifying how compound severity varies across phases and eviction strategies.

3 Background and Problem Setup

We formalize compound sparsity, define the Interaction Index, and introduce phase segmentation. Full mathematical details appear in Appendix A.

Compound sparsity. Let $s \in (0, 1)$ denote a unified sparsity budget. *Attention head pruning* removes the bottom-s fraction of heads per layer by Frobenius norm importance (Voita et al., 2019; Michel et al., 2019). *KV cache eviction* removes the bottom-s fraction of token positions from the cache. We compare two strategies: *importance-based*, which scores positions by position-normalized cumulative attention mass (Eq. 5 in Appendix A), and *random*, which evicts uniformly at random (Chen et al., 2024). Applying both simultaneously yields *compound sparsity*; the two axes interact because head pruning alters the importance ranking used for eviction (Liao et al., 2026). Existing joint-pruning approaches such as GRASPrune (Wang et al., 2026) learn global budget-constrained gates but treat decisions as static across generation.

Interaction Index. For a text segment, let PPL_c denote perplexity under condition $c \in \{d, a, k, b\}$ (dense, attention-only, KV-only, compound). Normalizing by the dense baseline yields $r_c = \text{PPL}_c/\text{PPL}_d$, and:

$$\text{II} = r_b - r_a - r_k + 1. \tag{1}$$

$\text{II} > 0$ indicates super-additive degradation; $\text{II} = 0$ indicates additive independence. We adopt the additive null because the two axes operate on disjoint dimensions; qualitative conclusions hold under the

multiplicative null (Appendix B). Because the II distribution is heavily right-skewed, we report medians throughout.

Severity gradient. To capture phase variation, the per-problem *amplification ratio* is:

$$G^{(i)} = \frac{\mathrm{II}^{(i)}_{low}}{\mathrm{II}^{(i)}_{high}}, \tag{2}$$

where $\mathrm{II}^{(i)}_{low/high}$ is the interaction index in the low/high-sparsity phase for problem i. We report $G = \widetilde{G^{(i)}}$, the median across problems, with bootstrap CIs. We use the median of per-problem ratios (not the ratio of medians) to preserve per-problem pairing; see Appendix D for comparison. $G \gg 1$ indicates phase-concentrated severity.

Phase segmentation. Inspired by ThinKV's phase-aware perspective on reasoning-time compression (Ramachandran et al., 2026), each generated sequence is partitioned into three phases (low-, mid-, high-sparsity) by tercile boundaries of per-position attention sparsity, using a quantile-based segmentation rather than explicit reasoning delimiters (e.g., `<think>` tokens). Low-sparsity phases correspond to broadly distributed attention (exploratory reasoning); high-sparsity phases to concentrated attention (structured computation). Details in Appendix A.

4 Experimental Setup

Models. We evaluate on two 7B-parameter models sharing the Qwen2 backbone (28 layers, 28 heads, 4-group GQA): **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI, 2025), distilled from DeepSeek-R1 with explicit chain-of-thought traces yielding pronounced phase structure; and **Qwen2.5-7B-Instruct** (Yang et al., 2024), with standard instruction tuning, providing a cross-model replication target. Both are loaded in `bfloat16` with eager attention.[1]

Dataset. We evaluate all 500 problems from MATH-500 (Hendrycks et al., 2021). Each problem is decoded greedily (`max_new_tokens`=2048). Problems with degenerate phase assignments (null low or high phase) are excluded, yielding effective sample sizes of n=496 for DeepSeek (4 excluded) and n=475 for Qwen (25 excluded). Qwen exhibits a higher mid-phase null rate (45.2% vs. 10.2% for DeepSeek) due to lower temporal autocorrelation in its attention sparsity patterns (lag-1 r=0.718 vs. 0.894, d= − 1.68); this does not affect the primary gradient metric $\Delta = \mathrm{II}_{low} - \mathrm{II}_{high}$, which requires only low and high phases. Sensitivity analysis across `min_phase_length` $\in \{5, 10, 20\}$ confirms robustness (Appendix L).

Sparsity configuration. We apply symmetric compound sparsity at $s \in \{0.15, 0.20, 0.25, 0.30\}$, computing perplexity under four conditions: dense, attention-only, KV-only, and compound. We compare *importance-based* eviction (position-normalized cumulative attention score, Eq. 5; BOS always retained) and *random* eviction (uniform). Heads are pruned per layer by Frobenius norm; under random eviction, both heads and positions are selected uniformly.

Phase segmentation and metrics. Each sequence is segmented into three phases using the quantile-based method (§3), with `min_phase_length`=10. We compute segment-level perplexity, II (Eq. 1), and G (Eq. 2), reporting medians due to heavy right-skew (mean/median ratios up to 43:1). Robustness to segmentation granularity (quartile, quintile; Appendix M) and minimum phase length (5, 10, 20 tokens; Appendix L) is verified.

Statistical analysis. All reported confidence intervals use nonparametric bootstrap (B=10,000, seed=42) (Efron and Tibshirani, 1993). Significance tests use Wilcoxon signed-rank (paired within-model) or Mann-Whitney U (cross-model), with Bonferroni correction at family-wise k=4 (per model×strategy combination, 4 sparsity levels testing one dose-response hypothesis). Benjamini-Hochberg FDR correction across all 44 tests is reported as a robustness check in Appendix K; all Bonferroni-significant results remain significant under BH-FDR. Dose-response monotonicity is confirmed by the Jonckheere-Terpstra test (z=6.2–33.7, all $p < 10^{-9}$), which uses per-problem values (∼496 per group) rather than aggregated medians.[2] Scaling from n=50 (profiling) to n=500 narrowed bootstrap CI widths by 2.5–5.9× across all 16 conditions, directly demonstrating precision gains. G

[1] An earlier `float16` configuration produced sporadic NaN values; `bfloat16` eliminated all occurrences.

[2] Spearman rank correlation on the four aggregate medians (ρ=1.0, exact permutation p=0.042, one-tailed) is consistent but has limited power at n=4 levels; we rely on the JT test as primary evidence.

Table 1: Median compound severity difference ($\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$) by eviction strategy and sparsity level on DeepSeek-R1-Distill-Qwen-7B (n=496, 500 MATH-500 problems).

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	0.39	1.21	3.99	8.48
Random	0.01	0.03	0.26	0.51

(amplification ratio, Eq. 2) is reported as an auxiliary metric; the paired median difference Δ is primary.[3]

5 Results

5.1 Strategy-Dependent Severity

Table 1 reports the median compound severity difference across strategies and sparsity levels. Under importance-based eviction, Δ increases from 0.39 at s=0.15 to 8.48 at s=0.30—a 22× amplification (all Bonferroni-corrected $p \ll 10^{-10}$). Random eviction develops a detectable but weak phase gradient above $s \approx 0.20$: at $s \leq 0.20$, Δ is non-significant (0.01–0.03; CIs span zero); at $s \geq 0.25$, Δ becomes significant (0.26–0.51; $p_{\text{Bonf}} < 0.006$) but 15–39× smaller than under importance-based eviction. Both strategies exhibit monotonic dose-response (Jonckheere-Terpstra z=6.2–33.7, all $p < 10^{-9}$), but importance masking effect sizes are an order of magnitude larger.

5.2 Phase-Modulated Amplification

Decomposing II by reasoning phase reveals that strategy-dependent severity is driven by amplification in the low-sparsity phase. Table 2 reports the per-phase breakdown at n=496. The per-problem amplification ratio G (Eq. 2) increases from 7.5× (95% CI: $[4.1, 11.4]$) at s=0.15 to 79.8× ($[63.3, 96.5]$) at s=0.30 under importance-based eviction, with all CIs excluding 1. Strong phase modulation is specific to importance-based eviction (G=7.5–79.8×). Random eviction shows weak but statistically significant modulation at $s \geq 0.25$ ($G \approx$ 1.1–1.3; CIs exclude 1), while at $s \leq 0.20$ the CIs span or touch 1. Figure 1 visualizes this contrast.

5.3 Phase-Strategy Crossover

For DeepSeek, a strategy ranking reversal emerges at all four sparsity levels (n=496): at s=0.25, importance-based eviction yields *lower* II than random in the high-sparsity phase (0.021 vs. 1.113) but *higher* II in the low-sparsity phase (3.971 vs. 1.514—a 2.6× reversal; both $p < 10^{-4}$), with reversal magnitude ranging from 2.2× to 4.0× across $s \in \{0.15, 0.20, 0.25, 0.30\}$. Qwen does not exhibit this crossover: importance-based eviction produces lower II than random in both phases (e.g., at s=0.25: high 0.004 vs. 1.108; low 1.053 vs. 1.769), indicating that the ranking reversal is model-dependent. Full crossover analysis appears in Appendix C.

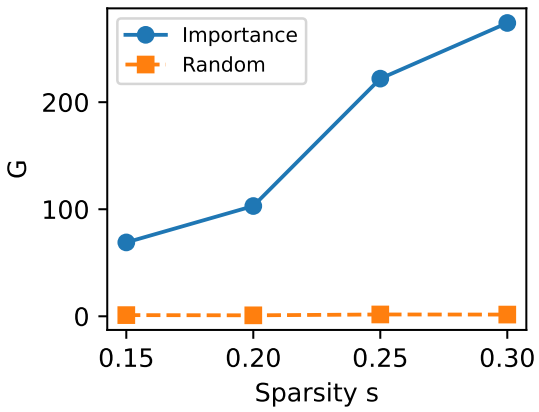


Figure 1: Amplification ratio (G) as a function of sparsity level s (profiling subset, n=50). Importance-based eviction (solid) exhibits an increasing gradient; random eviction (dashed) remains near 1×. At n=496, G ranges from 7.5× to 79.8× for importance and 0.5× to 1.3× for random (Table 2).

5.4 Replication Across Training Paradigms

DeepSeek-R1-Distill-Qwen-7B (distillation-trained) and Qwen2.5-7B-Instruct (instruction-tuned) share the Qwen2 backbone but differ in training objective, enabling controlled comparison of training paradigm effects. Qwen2.5-7B-Instruct (n_{eff}=475) reproduces the monotonic severity gradient: Δ increases from 0.07 at s=0.15 to 3.16 at s=0.30 (all Bonferroni-corrected $p < 10^{-14}$), paralleling DeepSeek at approximately one-quarter to one-sixth the magnitude. Both models yield strictly increasing median Δ trajectories (Jonckheere-Terpstra $z \geq 6.2$, $p < 10^{-9}$; see §4). Random eviction on Qwen mirrors DeepSeek: non-significant at $s \leq 0.20$, becoming significant at $s \geq 0.25$ (Δ=0.39–0.91; $p_{\text{Bonf}} < 10^{-10}$) but 2.7–20× weaker than importance-based. A caveat applies to mid-phase comparisons: on Qwen, 45.2% of problems yield null mid-phase segments

[3] Experiments on 4× NVIDIA RTX PRO 6000 Blackwell 96 GB GPUs.

Table 2: Per-phase interaction index (II) at n=496/475. Δ: paired median difference with bootstrap 95% CI (B=10,000). G: per-problem amplification ratio (Eq. 2) with bootstrap 95% CI. Profiling-subset values (n=50) in Table 5.

s	Strategy	$\widetilde{\text{II}}_{high}$	$\widetilde{\text{II}}_{low}$	Δ [95% CI]	G [95% CI]
DeepSeek-R1-Distill-Qwen-7B (n=496)					
0.15	Importance	0.003	0.357	0.39 [0.35, 0.44]	**7.5** [4.1, 11.4]
0.15	Random	0.083	0.090	0.01 [−0.02, 0.04]	0.5 [0.4, 0.6]
0.20	Importance	0.008	1.211	1.21 [1.11, 1.34]	**17.9** [12.7, 26.3]
0.20	Random	0.270	0.357	0.03 [−0.01, 0.08]	0.8 [0.6, 0.9]
0.25	Importance	0.021	3.971	3.99 [3.53, 4.36]	**49.1** [40.1, 57.8]
0.25	Random	1.113	1.514	0.26 [0.13, 0.35]	1.1 [1.0, 1.2]
0.30	Importance	0.038	8.456	8.48 [7.75, 9.69]	**79.8** [63.3, 96.5]
0.30	Random	3.112	3.848	0.51 [0.26, 0.89]	1.1 [1.0, 1.3]
Qwen2.5-7B-Instruct (n=475)					
0.15	Importance	−0.000‡	0.044	0.07 [0.05, 0.09]	n.d.†
0.15	Random	0.094	0.097	−0.01 [−0.04, 0.01]	0.7 [0.4, 0.8]
0.20	Importance	0.000	0.275	0.31 [0.26, 0.36]	**2.6** [0.5, 5.1]
0.20	Random	0.291	0.356	0.02 [−0.01, 0.07]	0.9 [0.7, 1.0]
0.25	Importance	0.004	1.053	1.07 [0.98, 1.18]	**9.5** [5.2, 14.8]
0.25	Random	1.108	1.769	0.39 [0.24, 0.55]	1.3 [1.2, 1.4]
0.30	Importance	0.014	3.208	3.16 [2.80, 3.56]	**36.9** [25.3, 49.1]
0.30	Random	2.819	4.253	0.91 [0.61, 1.26]	1.3 [1.2, 1.4]

† $\widetilde{\text{II}}_{high} \approx 0$; ratio undefined ($G$ CI: $[-0.2, 0.9]$). Δ=0.07, p=8.7×10^{-15} confirms $\text{II}_{low} > \text{II}_{high}$. ‡ Negative II at this sparsity reflects `bfloat16` numerical noise when the interaction effect approaches zero ($|\text{II}| < 10^{-4}$).

due to low temporal autocorrelation in attention sparsity (lag-1 r=0.718 vs. DeepSeek's 0.894; d= − 1.68), preventing formation of contiguous mid-phase runs above the `min_phase_length`=10 threshold. This does not affect the primary low–high contrast (Δ). Per-phase effective sample sizes are reported in Appendix E. Full cross-model tables appear in Appendix E.

5.5 Exploratory Cross-Architecture Analysis

To test whether the phase-modulated severity pattern extends beyond the Qwen2 architecture, we replicate the experiment on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, 8B parameters; n=49 paired problems). Table 3 reports the strategy-dependent severity comparison across sparsity levels.

Table 3 reports the full results. The evidence for a strategy contrast on LLaMA is mixed: at s=0.15, the bootstrap CI excludes zero ($[0.002, 0.043]$) but the Bonferroni-corrected Wilcoxon test is non-significant (p_{Bonf}=0.219); conversely, at s=0.30 the CI spans zero ($[-0.069, 0.544]$) while the Wilcoxon test reaches significance (p_{Bonf}=0.029). At intermediate levels ($s \in \{0.20, 0.25\}$), neither CI nor Wilcoxon reaches significance. The median paired difference increases monotonically ($\widetilde{\Delta}$: 0.020 → 0.356), consistent with a genuine but weak trend—partially confirming the cross-architecture hypothesis while underscoring the substantially weaker interaction on LlamaForCausalLM.[4]

The absolute magnitude is architecture-dependent: LLaMA severity differences are ∼40× smaller than DeepSeek and ∼7× smaller than Qwen. Notably, random eviction on LLaMA also exhibits a non-trivial phase gradient (d_z: 0.35 → 0.44), substantially reducing the strategy contrast that is the hallmark of the phase-strategy interaction on Qwen2 models. This suggests that the strength of the strategy×phase interaction—not merely the severity gradient itself—is architecture-dependent. The directional pattern—importance producing larger phase gradients than random—is qualitatively consistent across all three models, but the LLaMA evidence should be interpreted as preliminary given the weak interaction magnitude and partial statistical significance.

[4]The CI/Wilcoxon discrepancy reflects differing power profiles under small n with heavy-tailed distributions: the percentile bootstrap CI is based on the resampled median distribution and is less sensitive to tail observations, while the Wilcoxon signed-rank test uses signed ranks and is more sensitive to tail outliers. This inconsistency itself supports the conservative "mixed/weak evidence" interpretation.

Table 3: Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM). $\widetilde{\Delta}$: median paired difference (importance − random) with bootstrap 95% CI (B=10,000); p: Bonferroni-corrected Wilcoxon signed-rank p (k=4). n=49 paired problems.

s	$\widetilde{\Delta}$ [95% CI]	p_{Bonf}
0.15	0.020 [0.002, 0.043]	0.219
0.20	0.022 [−0.090, 0.187]	0.253
0.25	0.095 [−0.004, 0.303]	0.083
0.30	0.356 [−0.069, 0.544]	**0.029**

Table 4: Paired median difference ($\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$) under importance-based eviction with bootstrap 95% CIs (B=10,000).

s	DS (n=496): $\widetilde{\Delta}$ [95% CI]	QW (n=475): $\widetilde{\Delta}$ [95% CI]
0.15	0.39 [0.35, 0.44]	0.07 [0.05, 0.09]
0.20	1.21 [1.11, 1.34]	0.31 [0.26, 0.36]
0.25	3.99 [3.53, 4.36]	1.07 [0.98, 1.18]
0.30	8.48 [7.75, 9.69]	3.16 [2.78, 3.56]

All $p_{\text{Bonf}} < 10^{-13}$. Within-importance d_z (s015 vs. s030): DS 0.94, QW 0.92.

5.6 Statistical Validation

Importance-based eviction produces significantly larger phase-dependent severity than random eviction at all four sparsity levels: the median paired difference (importance − random) increases from 0.42 at s=0.15 to 9.26 at s=0.30 (all $p_{\text{Bonf}} \ll 10^{-10}$, Wilcoxon signed-rank). Full permutation test results from the n=50 profiling subset appear in Appendix F.

Table 4 reports the paired median difference Δ under importance-based eviction for both models at n=496/475. All eight conditions yield CIs excluding zero. The within-importance dose-response d_z (s015 vs. s030) is 0.94 for DeepSeek and 0.92 for Qwen, both large effects.[5]

We provide a correlational mechanistic analysis in terms of discrimination collapse in §6.

6 Discrimination Collapse under Compound Sparsity

We present discrimination collapse as a *correlational diagnostic* rather than a complete causal explanation. The consistent Gini decrease across all conditions provides a necessary background—aggregate attention concentration drops uniformly under head pruning regardless of eviction strategy (importance ΔGini: -0.030 to -0.081; random ΔGini: -0.031 to -0.071; $<10\%$ difference). However, severity diverges by 15–17× between strategies despite near-identical Gini changes, indicating that concentration collapse alone is insufficient. The critical differentiator is *structural specificity*: importance-based pruning systematically removes high-attention-score heads, disrupting the attention pattern structure that KV eviction relies on, while random pruning preserves more structural diversity. The quantitative gap between score-level changes (6–16% relative Gini reduction from a DeepSeek baseline of $\widetilde{\text{Gini}} \approx 0.50$; Qwen baseline ≈ 0.57) and outcome-level amplification (G=7.5–79.8×) indicates additional nonlinear factors (e.g., cascading effects across layers) that warrant dedicated investigation.

6.1 Discrimination Collapse

Head pruning systematically reduces the discriminative power of cumulative importance scores, suggesting a mechanism for the phase-strategy interaction. The Gini coefficient decreases by 0.030–0.081 (DeepSeek, 6–17% relative to dense baseline) and 0.010–0.020 (Qwen, 2–4%) after pruning, with consistent negative deltas in all 16 conditions (Appendix G, Table 9). Critically, random eviction shows near-identical Gini drops at matched sparsity (DeepSeek s=0.25: importance -0.065 vs. random -0.060; s=0.30: -0.081 vs. -0.071; Qwen: -0.019 to -0.046 for both strategies), confirming that discrimination collapse is a consequence of head pruning *per se*, not the eviction strategy. Yet importance-based severity is 15–17× larger than random at matched sparsity—the amplification arises because importance-based eviction *depends on* the very scores that pruning degrades, while random eviction's decisions are score-independent.

This pattern—which we term *discrimination collapse*—is consistent with importance-based eviction degenerating into noise-sensitive selection when score distributions flatten. The effect is phase-dependent: in low-sparsity phases, attention is already broadly distributed, so pruning-induced flattening has maximum impact; in high-sparsity phases, concentration is preserved. This observation is independently corroborated by (Choi et al., 2025), who report importance score collapse under heavy single-axis pruning; our results extend this to the compound setting, where collapse is amplified by the interaction of two compression axes and modulated by reasoning phase.

[5] At n=50 profiling, per-condition d_z ranged from 0.44–0.80 (DS) and 0.28–1.02 (QW); d_z instability at high sparsity due to outlier-driven variance resolved at n=500.

We note that this evidence is correlational; causal validation through controlled manipulation of importance-score entropy is an important direction for future work.

6.2 Entropy Redistribution

As a supplementary observation, head pruning also redistributes the attention entropy of surviving heads on DeepSeek ($\Delta\mathcal{H} \in [-0.047, -0.040]$; Appendix G, Table 10). However, entropy redistribution shows model-dependent behavior—the direction reverses on Qwen at s=0.25 ($\Delta\mathcal{H} = +0.045$)—and is not a universal mechanism. Discrimination collapse remains the robust, model-independent correlate.

7 Discussion

Quantitative surprise. While the qualitative direction—importance scores degrade when attention is diffuse—may seem expected, the quantitative structure is not predictable from first principles. The severity gradient (G=7.5–79.8× at n=496) far exceeds a naive "somewhat worse" prediction. The crossover on DeepSeek—where importance-based eviction simultaneously outperforms random in high-sparsity phases ($\widetilde{\mathrm{II}} = 0.021$ vs. 1.113) and underperforms in low-sparsity phases (3.971 vs. 1.514)—cannot be derived from single-axis entropy reasoning.

Implications for deployment. Applying a uniform compression budget across all inference steps implicitly accepts order-of-magnitude severity penalties during low-sparsity reasoning phases. A *phase-aware compression budget*—allocating lower sparsity to phases with $G \gg 1$ and reclaiming budget in near-additive high-sparsity phases—could mitigate this interaction while preserving compression gains. The severity gradient provides a natural signal for such allocation. We expect qualitatively similar behavior for other attention-mass-based scoring methods (H_2O (Zhang et al., 2023), SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2025b)), as they share the property that head pruning alters the score distribution feeding into eviction decisions; however, quantitative magnitudes may differ given implementation-specific scoring mechanisms.

Characterization as diagnostic signal. The primary value of the phase-dependent severity gradient is diagnostic rather than prescriptive: it identifies *where* compound sparsity is most destructive within a generation. The monotonically increasing G across sparsity levels provides a natural signal for allocating compression budgets—phases with $G \gg 1$ are precisely those where uniform budgets incur order-of-magnitude penalties. The pilot experiment below confirms the direction is correct (Wilcoxon-significant recovery) but the effect size is insufficient under coarse phase boundaries, suggesting that effective mitigation requires finer-grained phase-aware optimization. We position mitigation as an open problem for which our characterization provides the necessary foundation: future work can leverage the severity gradient as an objective for joint optimization of budget allocation and strategy selection.

A preliminary pilot experiment explored oracle phase-aware budget reallocation, shifting sparsity budget from high-severity low-sparsity phases to low-severity high-sparsity phases. At s=0.25, oracle recovery—the fraction of compound interaction eliminated by phase-aware allocation—reached 18% (Wilcoxon $p = 0.002$), below the pre-registered 30% gate for claiming practical mitigation. A heuristic variant achieved 32% recovery ($p = 0.024$), but 95% CIs for both conditions span zero.

Extended discussion in Appendix I.

Architecture-dependent magnitude. Cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM) provides partial support for the qualitative pattern—monotonically increasing severity gradient under importance-based eviction with within-strategy phase gradient $d_z \geq 1.0$—while revealing that absolute severity magnitudes are architecture-dependent (∼40× smaller than DeepSeek, ∼7× smaller than Qwen) and the strategy–phase interaction is substantially weaker than in Qwen2-based architectures. Notably, random pruning also exhibits a non-trivial phase gradient on LLaMA (d_z: 0.35 → 0.44), reducing the contrast between strategies.

8 Conclusion

We showed that compound sparsity severity—the super-additive interaction between head pruning and KV cache eviction—is not phase-invariant but exhibits an emergent phase-strategy interaction spanning over an order of magnitude within a single generation (n=496/475 problems). Under importance-based eviction, the amplification ratio G ranges from 7.5× to 79.8× across spar-

sity levels, with paired median differences Δ from 0.39 to 8.48 (DeepSeek) and 0.07 to 3.16 (Qwen), all $p < 10^{-13}$. Random eviction develops a detectable phase gradient only above $s \approx 0.20$, but 3–39$\times$ weaker than importance-based. A crossover on DeepSeek at s=0.25—where importance-based eviction reverses ranking across phases (2.6$\times$ reversal)—demonstrates that no single strategy dominates uniformly; Qwen does not exhibit this crossover, indicating model dependence. The pattern replicates on Qwen2.5-7B-Instruct at all four sparsity levels; suggestive preliminary evidence from DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM) indicates the phenomenon may extend beyond Qwen2, though at $\sim 1/40$ the magnitude and with a weaker strategy contrast. Discrimination collapse provides a correlational diagnostic supported by both Qwen2-based models, with structural specificity of importance-based pruning as the key differentiator from random eviction—despite near-identical Gini drops ($< 10\%$ difference), severity diverges by 15–17$\times$. These findings establish the phase-dependent nature of compound severity as a prerequisite and diagnostic foundation for future phase-aware mitigation strategies. Extending this analysis to code reasoning (HumanEval, LiveCodeBench) and science reasoning (GPQA), validating the discrimination collapse mechanism through causal intervention, and characterizing asymmetric compound interactions are promising directions for future work.

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Limitations

Model scale and architecture. All experiments are conducted at the 7–8B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Preliminary cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, §5.5) partially

confirms the qualitative pattern beyond the Qwen2 architecture—monotonically increasing d_z with within-importance phase gradient $\sim$1.0—but with substantially weaker interaction magnitude ($\sim$1/40 of DeepSeek) and mixed statistical evidence (only s=0.30 significant after Bonferroni correction). The pattern has been tested on two architecture families (Qwen2ForCausalLM, LlamaForCausalLM); comprehensive validation across more diverse architectures (e.g., Mistral, multi-latent attention) and larger scales is needed for full generalization claims.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. MATH-500 is a suitable first testbed because it elicits structured reasoning with clear phase transitions (exploratory vs. structured computation), supports per-problem evaluation enabling paired statistical tests, and spans sufficient difficulty to produce meaningful variation in sequence length and attention dynamics. However, different reasoning tasks—code generation (e.g., HumanEval, LiveCodeBench), science reasoning (e.g., GPQA), multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Whether the phase-strategy interaction documented here extends to these domains is an open question.

Symmetric sparsity assumption. Our analysis applies symmetric sparsity rates: the same fraction s is used for both head pruning and KV cache eviction. Practical deployments may use asymmetric budgets (e.g., aggressive head pruning with conservative cache eviction, or vice versa). The interaction dynamics under asymmetric compound sparsity may differ from the symmetric case studied here; investigating asymmetric compound interactions is a natural extension.

Mechanism. The discrimination collapse mechanism is supported by correlational evidence (consistent Gini/CV reduction across all conditions); causal validation through controlled entropy manipulation is needed. The entropy redistribution component replicates on only 2/4 Qwen conditions, suggesting model-specific factors.

Perplexity vs. accuracy. Our severity metric is perplexity-based (II), which measures next-token prediction degradation and has a known nonlinear relationship with downstream accuracy. We chose perplexity because its per-token granularity enables the phase-level decomposition central to our analysis; accuracy is binary per-problem and cannot be meaningfully decomposed by reasoning phase. Future work should investigate problem-level accuracy regression on phase-specific II to quantify how the perplexity-level severity gradient translates to task performance.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics. We do not report inference throughput or latency measurements, which are necessary for a complete deployment analysis but orthogonal to our severity characterization. Extended limitations in Appendix J.

Reproducibility

Code and data will be released upon acceptance.

A Compound Sparsity Formulation

Attention head pruning. Heads are ranked by Frobenius norm importance $w_j^{(\ell)} = \|\mathbf{A}_j^{(\ell)}\|_F$ (Voita et al., 2019; Michel et al., 2019); a binary mask $m_j^{(\ell)} \in \{0, 1\}$ retains the top-$(1-s) \cdot H$ heads:

$$\tilde{\mathbf{o}}_j^{(\ell)} = m_j^{(\ell)} \cdot \mathbf{o}_j^{(\ell)}. \tag{3}$$

KV cache eviction. A position mask $\mathbf{m}_k \in \{0, 1\}^T$ retains the top-$(1-s) \cdot T$ positions (BOS always preserved):

$$\tilde{a}_{ij} = a_{ij} \cdot m_{k,j}, \quad \forall\, i, j. \tag{4}$$

Under importance-based eviction, each position j is scored by its cumulative attention mass with position normalization:

$$\text{imp}(j) = \frac{\sum_{\ell,h,i} a_{ij}^{(\ell,h)}}{T - j}, \tag{5}$$

where the denominator compensates for causal bias—earlier positions are attended to by more queries (Ramachandran et al., 2026; Feng et al., 2026). Under random eviction, positions (excluding BOS) are evicted uniformly at random.

Compound sparsity. Applying both mechanisms simultaneously yields compound sparsity. The two axes interact: head pruning redistributes attention mass, altering the importance ranking for KV eviction. Prior work (Liao et al., 2026) reports super-additive joint effects but treats interaction magnitude as a fixed property of s.

Attention sparsity metric. For each query position i and head h, we compute the fraction of key positions receiving negligible attention:

$$\sigma_i^{(h)} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{1}\left[a_{ij}^{(h)} < 0.01 \cdot \max_k a_{ik}^{(h)} \right]. \quad (6)$$

The per-position sparsity $\sigma_i = \frac{1}{H} \sum_h \sigma_i^{(h)}$ averages across all heads. Tokens are partitioned into high-, mid-, and low-sparsity phases by the tercile boundaries of $\{\sigma_i\}$; segments shorter than 10 tokens are merged into adjacent phases.

B Multiplicative Null Verification

We verified that replacing the additive null $r_b = r_a + r_k - 1$ (Eq. 1) with the multiplicative null $r_b = r_a \cdot r_k$ does not change qualitative conclusions. Under the multiplicative formulation, the interaction index becomes $\mathrm{II}_\times = r_b - r_a \cdot r_k$. Across all sparsity levels and both eviction strategies, the sign and relative ordering of $\mathrm{II}_\times$ match those of the additive II: importance-based eviction exhibits a monotonically increasing severity gradient ($G_\times = 58$–$241\times$), while random eviction remains flat ($G_\times \approx 1$). The crossover at s=0.25 is preserved. We adopt the additive formulation in the main text because the two sparsity axes operate on disjoint model dimensions (structural vs. positional), making additive independence the more natural null hypothesis at the sparsity levels studied.

C Phase-Strategy Crossover Details

On DeepSeek (n=496), a strategy ranking reversal emerges at s=0.25 (Figure 2): importance-based eviction yields *lower* II than random in the high-sparsity phase (0.021 vs. 1.113) but *higher* II in the low-sparsity phase (3.971 vs. 1.514—a 2.6× reversal; both $p < 10^{-4}$). This crossover is confirmed at all four sparsity levels, with reversal magnitude of 4.0× at s=0.15, 3.4× at s=0.20, 2.6× at s=0.25, and 2.2× at s=0.30 (decreasing as random eviction severity rises). The reversal arises because head pruning redistributes attention mass, altering importance rankings in a way that couples the two sparsity mechanisms in low-sparsity phases (§G.1). Qwen does not exhibit a crossover: importance-based eviction produces lower II than random in both phases across all sparsity levels (e.g., at s=0.25: high 0.004 vs. 1.108; low 1.053 vs. 1.769).

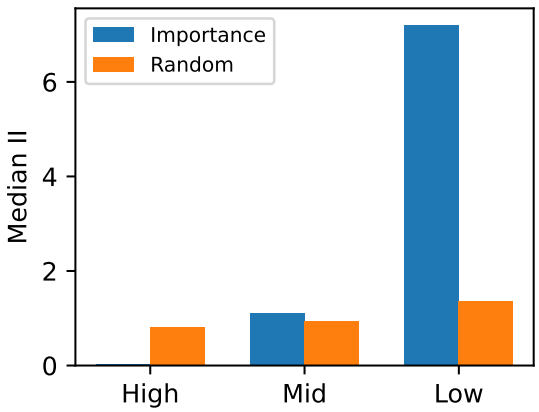


Figure 2: Strategy-dependent crossover at s=0.25 on DeepSeek (n=496). Grouped bar chart of median II by eviction strategy and reasoning phase. Importance-based eviction outperforms random in the high-sparsity phase but produces 2.6× higher II in the low-sparsity phase, reversing the strategy ranking.

D Profiling Subset Per-Phase Breakdown

E Cross-Model Replication Details

Both models share the Qwen2ForCausalLM backbone (28 layers, 28 heads, 4-group GQA) but differ in training: DeepSeek uses reasoning distillation from DeepSeek-R1, while Qwen2.5 uses standard instruction tuning without explicit reasoning delimiters. This controlled comparison isolates the effect of training methodology on the compound sparsity interaction.

Despite lacking explicit `<think>` delimiters, Qwen2.5-7B-Instruct exhibits sufficient variation in attention sparsity to permit phase segmentation via the same quantile-based method (§3). Twenty-five of the 500 problems are excluded due to degenerate phase assignments, yielding n=475 valid problems for Qwen (vs. n=496 for DeepSeek).

Per-phase effective sample size. Table 6 reports the per-phase effective sample size for Qwen2.5-7B-Instruct, defined as the number of problems with non-null phase segments (segment length $\geq$ `min_phase_length`). The mid-phase null rate (45.2%) reflects problems where mid-sparsity tokens do not form contiguous runs above the merging threshold; this does not affect the low–high contrast (Δ) used in the primary analysis.

Table 7 reports the cross-model comparison. Qwen reproduces the monotonic severity gradient at $\sim 1/5$ of DeepSeek's magnitude. Figure 3 visualizes this comparison.

Table 5: Per-phase interaction index (II) from the n=50 profiling subset (DeepSeek-R1-Distill-Qwen-7B). The main analysis (n=496) appears in Table 2. G values in this table use the ratio-of-medians definition ($\widetilde{\text{II}}_{low}/\widetilde{\text{II}}_{high}$), which differs from the primary per-problem definition in Eq. 2. At low sparsity (s=0.15), the two definitions diverge substantially (7.5× vs. 69×) because per-problem ratios are sensitive to near-zero II_{high} values.

s	Strategy	$\widetilde{\text{II}}_{high}$	$\widetilde{\text{II}}_{mid}$	$\widetilde{\text{II}}_{low}$	Δ	G
0.15	Importance	0.006	0.081	0.398	0.39	**69**
0.15	Random	0.093	0.097	0.103	0.01	1.1
0.20	Importance	0.015	0.203	1.496	1.48	**103**
0.20	Random	0.292	0.350	0.276	−0.02	0.9
0.25	Importance	0.032	1.118	7.196	7.16	**222**
0.25	Random	0.818	0.932	1.363	0.54	1.7
0.30	Importance	0.053	2.283	14.639	14.59	**274**
0.30	Random	2.503	2.592	4.041	1.54	1.6

Table 6: Per-phase effective sample size for Qwen2.5-7B-Instruct ($n_{\text{eff}} = 475$). Phase segmentation is performed once on dense-model attention patterns and is independent of sparsity level.

Phase	s=0.15	s=0.20	s=0.25	s=0.30
High	475	475	475	475
Mid	274	274	274	274
Low	475	475	475	475

Table 7: Cross-model comparison of median compound severity difference (Δ) under importance-based eviction.

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-Distill-Qwen-7B (n=496)	0.39	1.21	3.99	8.48
Qwen2.5-7B-Instruct (n=475)	0.07	0.31	1.07	3.16

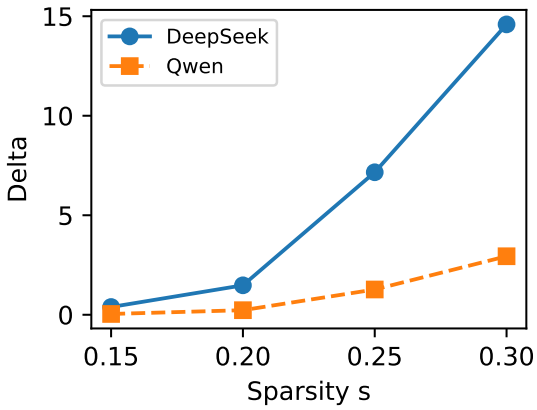


Figure 3: Cross-model comparison of the median severity difference under importance-based eviction (n=496/475). Both DeepSeek (solid) and Qwen (dashed) exhibit monotonically increasing Δ with sparsity; Qwen's magnitude is approximately one-quarter to one-sixth of DeepSeek's.

Per-problem concordance. The per-problem concordance across models is not significant (mean $\rho = -0.107$, $p = 0.926$), indicating that while the aggregate gradient direction replicates, the specific problems driving severity differ between models—consistent with model-specific attention patterns rather than a universal per-problem vulnerability ranking.

Random eviction control. As a control, random eviction on Qwen mirrors DeepSeek: non-significant at $s \leq 0.20$ (Δ: −0.01, 0.02; CIs span zero) but significant at $s \geq 0.25$ (Δ=0.39, $p_{\text{Bonf}} < 10^{-10}$; Δ=0.91, $p_{\text{Bonf}} < 10^{-13}$). Where significant, the random phase gradient is 2.7–3.5× weaker than importance-based (0.39 vs. 1.07 at s=0.25; 0.91 vs. 3.16 at s=0.30), confirming strategy-specific amplification.

F Permutation Test Details

For each problem i and sparsity level s, we compute the compound severity difference $\Delta_i^{(\text{strat})} = \text{II}_{low,i} - \text{II}_{high,i}$ under each eviction strategy. The test statistic is the median of the paired difference $\widetilde{\Delta} = \text{median}_i\big(\Delta_i^{(\text{imp})} - \Delta_i^{(\text{rand})}\big)$, measuring whether importance-based eviction produces systematically larger phase-dependent severity than random eviction. We permute strategy labels 10,000 times to construct the null distribution and apply Bonferroni correction for 4 sparsity-level comparisons.

Per-problem distributions of the paired severity difference $\Delta_i = \text{II}_{low,i} - \text{II}_{high,i}$ under importance-based eviction are right-skewed at all sparsity levels, with mean/median ratios ranging from 1.2×

Table 8: Paired permutation test for compound severity: importance vs. random eviction on DeepSeek-R1-Distill-Qwen-7B (n=49 profiling subset, 10,000 permutations, Bonferroni-corrected). The monotonic Δ pattern is confirmed at n=496 (Table 1).

s	Observed $\widetilde{\Delta}$	95% CI	p_{Bonf}
0.15	0.37	[0.19, 0.62]	0.0032
0.20	0.97	[0.69, 1.49]	< 0.001
0.25	3.34	[1.66, 6.64]	< 0.001
0.30	9.85	[6.70, 16.29]	< 0.001

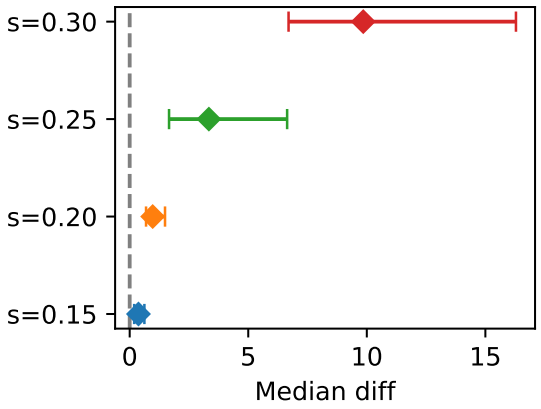


Figure 4: Forest plot of permutation test results. Horizontal bars show 95% permutation CIs; diamonds mark observed median differences $\widetilde{\Delta}$. All CIs lie entirely above zero, confirming significant strategy-dependent severity at every sparsity level.

(s=0.15) to 1.6× (s=0.30) for DeepSeek and 1.1× to 1.4× for Qwen.

The permutation null distributions are approximately symmetric around zero at all sparsity levels. At s=0.15, the observed statistic falls at the 99.92th percentile of the null ($p_{\text{Bonf}} = 0.0032$); at $s \geq 0.20$, no permutation exceeds the observed statistic.

Table 8 reports the results. All four sparsity levels reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30. The monotonic growth of both the point estimate and the CI lower bound ($0.19 \rightarrow 0.69 \rightarrow 1.66 \rightarrow 6.70$) confirms that compound severity amplification strengthens with sparsity—not merely in central tendency but also in worst-case lower bounds. Figure 4 presents these results as a forest plot.

Table 9: Discrimination collapse under importance-based eviction. ΔGini and ΔCV are post-pruning minus baseline; negative values indicate reduced discriminative power.

Model	s	ΔGini	ΔCV
DeepSeek	0.15	−0.030	−0.56
DeepSeek	0.20	−0.040	−0.77
DeepSeek	0.25	−0.065	−1.29
DeepSeek	0.30	−0.081	−1.66
Qwen	0.15	−0.010	−0.05
Qwen	0.20	−0.012	−0.21
Qwen	0.25	−0.010	−1.49
Qwen	0.30	−0.020	−1.14

G Analysis: Entropy-Discrimination Collapse

G.1 Discrimination Collapse

Head pruning reduces the number of heads contributing to cumulative importance scores, flattening the score distribution. Table 9 quantifies this effect: the Gini coefficient decreases by 0.030–0.081 (DeepSeek) and 0.010–0.020 (Qwen) after pruning, with consistent negative deltas across all 16 conditions (both models × 4 sparsity levels × 2 strategies). The CV delta is likewise negative in all 8 importance-based conditions. The same pattern holds under random eviction, confirming that discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

When importance scores lose discriminability, thresholding degenerates into noise-sensitive selection—functionally approaching biased random eviction. This is phase-dependent: in low-sparsity phases, attention is already broadly distributed, so pruning-induced flattening has maximum impact; in high-sparsity phases, concentration is preserved. Random eviction is immune because its decisions are independent of score quality, explaining $G \approx 1$.

G.2 Entropy Redistribution

Beyond discrimination collapse, we examine whether head pruning also redistributes the attention entropy of surviving heads. We measure the mean attention entropy per head:

$$\mathcal{H}_i^{(h)} = -\sum_{j=1}^{T} a_{ij}^{(h)} \log a_{ij}^{(h)} \tag{7}$$

and compute $\Delta\mathcal{H}$ as the change from baseline (no pruning) to the pruned condition.

Table 10: Attention entropy deltas ($\Delta\mathcal{H}$) under importance-based eviction. DeepSeek shows consistent entropy compression; Qwen partially replicates with a reversal at s=0.25.

Model	s	$\Delta\mathcal{H}$
DeepSeek	0.15	-0.040
DeepSeek	0.20	-0.047
DeepSeek	0.25	-0.045
DeepSeek	0.30	-0.044
Qwen	0.15	-0.073
Qwen	0.20	-0.076
Qwen	0.25	$+0.045$
Qwen	0.30	-0.007

For DeepSeek under importance-based eviction, the entropy delta is consistently negative across all four sparsity levels ($\Delta\mathcal{H} \in [-0.047, -0.040]$; Table 10), indicating that importance-based pruning systematically selects heads whose removal concentrates attention in the surviving heads, reinforcing discrimination collapse.

For Qwen, the entropy pattern only partially replicates. The delta is negative at s=0.15 ($\Delta\mathcal{H} = -0.073$) and s=0.20 (-0.076), consistent with DeepSeek. However, at s=0.25, the delta reverses to $+0.045$, with 22 of 30 problems showing positive entropy change. At s=0.30, the delta is near zero (-0.007). The s=0.25 reversal is systematic rather than a statistical artifact: it is consistent across all three reasoning phases (low: $+0.042$, mid: $+0.034$, high: $+0.017$). We interpret this as a genuine model-specific difference, possibly arising from distinct head specialization patterns: Qwen's instruction tuning (vs. DeepSeek's distillation-based training) may produce head configurations where removing 25% of heads triggers compensatory broadening rather than concentration.

We characterize discrimination collapse as the robust, model-independent driver of the phase-strategy interaction, with entropy redistribution as a secondary amplifier that is consistent in DeepSeek but model-dependent in Qwen.

H Discrimination Collapse Details

This cross-model consistency (Table 9) establishes discrimination collapse as a robust consequence of compound sparsity. When importance scores lose discriminative power, thresholding the bottom-s fraction becomes noise-sensitive: small perturbations change which positions cross the eviction boundary. In the limit of perfectly uniform scores, importance-based eviction degenerates into an arbitrary selection functionally equivalent to biased random eviction. The same pattern holds under random eviction (DeepSeek: -0.031 to -0.071; Qwen: -0.019 to -0.046), confirming that discrimination collapse is a consequence of head pruning *per se*, not of the eviction strategy.

I Extended Discussion

On the predictability of phase-modulated severity. The compound interaction is not uniformly fragile—it is selectively fragile in a phase-dependent manner ($\widetilde{\Pi}_{\text{high}} < 0.06$ at $s \leq 0.30$).

Random masking as falsification control. At n=500, random eviction develops a detectable but weak phase gradient above $s \approx 0.20$: non-significant at $s \leq 0.20$ for both models, but significant at $s \geq 0.25$ (DS: Δ=0.26–0.51, $p_{\text{Bonf}} < 0.006$; QW: Δ=0.39–0.91, $p_{\text{Bonf}} < 10^{-10}$). Where significant, the random phase gradient is 3–39$\times$ weaker than importance-based, confirming that the dominant amplification is strategy-specific.

Entropy mechanism boundary conditions. Directly measuring per-head entropy shifts across architecturally distinct models—and identifying what training-time properties determine whether entropy concentrates or disperses under head pruning—would clarify the boundary conditions of the full mechanism.

J Extended Limitations

Model scale. All experiments are conducted at the 7B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Larger models may exhibit different attention redundancy structures that alter the discrimination collapse dynamics (and the model-dependent entropy component may behave differently at scale).

Architectural scope. Our two models—DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct—share the Qwen2ForCausalLM architecture. The cross-model replication (§5.4) establishes that the findings are not artifacts of a single model's weights, though both models share the same Qwen2ForCausalLM architecture. Preliminary cross-architecture validation on DeepSeek-R1-Distill-Llama-8B (LlamaForCausalLM, §5.5) shows qualitatively consistent within-importance phase gradients ($d_z \sim 1.0$) but

a substantially weaker strategy × phase interaction than in Qwen2-based architectures; random pruning also exhibits a non-trivial phase gradient on LLaMA, reducing the strategy contrast. This partial confirmation (s=0.30 Wilcoxon p_{Bonf}=0.029; $s \leq 0.25$ non-significant) suggests the qualitative pattern may generalize but quantitative impact is architecture-dependent. Comprehensive cross-architecture validation across more diverse designs (e.g., Mistral, multi-latent attention architectures) and scales remains an important next step.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Generalizing the phase-strategy interaction to broader reasoning domains is an important next step.

Statistical scope. Bootstrap confidence intervals use B=10,000 resamples (seed=42). The primary analysis uses n=496 paired problems for DeepSeek (4 excluded) and n=475 for Qwen (25 excluded); the permutation test (Table 8) uses n=49 from an initial profiling subset. The severity difference distribution is right-skewed (mean/median ratios up to 43:1), motivating our use of medians and nonparametric tests rather than parametric alternatives. All four profiling-subset permutation tests reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$; Table 8), with CI lower bounds increasing from 0.19 to 6.70—confirming the effect direction at every sparsity level. At n=496, Wilcoxon signed-rank tests confirm all conditions with $p_{\text{Bonf}} < 10^{-13}$ (Table 4). The permutation CIs at higher sparsity are wider (e.g., $[6.70,\ 16.29]$ at s=0.30), reflecting genuine variance in per-problem severity amplification.

At n=500, random eviction on both models becomes significant at $s \geq 0.25$ (Qwen: Δ=0.39, $p_{\text{Bonf}} < 10^{-10}$ at s=0.25; Δ=0.91, $p_{\text{Bonf}} < 10^{-13}$ at s=0.30; bootstrap CIs exclude zero). These effects are genuine but 2.7–20× weaker than importance-based eviction at the same sparsity levels, consistent with a weak phase gradient that is detectable at sufficient sample size but not the primary driver of compound severity.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$, a moderate range representative of practical deployment targets. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics—the sub-exponential growth pattern suggests the severity gradient may plateau, but this remains unverified. Conversely, the lower boundary at which phase-modulated amplification first emerges is not yet characterized.

Table 11: Bonferroni vs. BH-FDR correction summary (α=0.05, 44 tests). "BH only" denotes tests significant under BH-FDR but not Bonferroni.

Category	Bonf. sig.	BH-only sig.
Per-condition Wilcoxon (k=16)	10	2
Imp vs. rand (k=8)	8	0
Adjacent sparsity (k=12)	10	1
Cross-model Mann-Whitney (k=8)	4	2
Total (k=44)	32	5

No throughput evaluation. We do not report inference throughput or latency, which are necessary for deployment analysis but orthogonal to our severity characterization.

K Benjamini-Hochberg FDR Correction

Table 11 compares Bonferroni (k=4, per model×strategy family) and Benjamini-Hochberg FDR correction across all 44 statistical tests. BH-FDR recovers 5 additional significant results beyond Bonferroni, all in low-effect-size conditions (random eviction or cross-model random comparisons). All 32 Bonferroni-significant results remain significant under BH-FDR.

The 5 BH-only results are: DeepSeek random s=0.25 and s=0.30 per-condition Wilcoxon (p_{Bonf}=0.064/0.065, BH q=0.002), DeepSeek random s=0.25 → 0.30 adjacent-sparsity step (p_{Bonf}=0.078, BH q=0.002), and cross-model random s=0.25/s=0.30 Mann-Whitney (p_{Bonf}=0.66/0.17, BH q=0.018/0.005). All involve random eviction—the weakest effects—confirming that Bonferroni is conservative but does not miss any importance-based findings.

L Minimum Phase Length Sensitivity

Table 12 reports the paired median difference Δ under three `min_phase_length` (mpl) settings across all 16 conditions (2 models × 2 strategies × 4 sparsity levels). Conclusions are robust: all importance-based conditions remain significant at $p_{\text{Bonf}} < 10^{-10}$ across mpl $\in \{5, 10, 20\}$; random condi-

tions maintain the same significance pattern (non-significant at $s \leq 0.20$, significant at $s \geq 0.25$). Median Δ values vary by $< 5\%$ between mpl settings for DeepSeek importance, confirming that the severity gradient is not an artifact of the merging threshold.

The main effect of increasing mpl is reduced n_{eff}: at mpl=20, DeepSeek retains 433/500 problems and Qwen retains 289/500 (mid-null rate rises to 71.2% and 84.6%, respectively). Despite this sample reduction, all significance conclusions hold.

M Segmentation Robustness

Table 13 reports the paired median difference Δ under tercile, quartile, and quintile segmentation for both models under importance-based eviction. Phase gradient direction and statistical significance are preserved under quartile and quintile segmentation (n=50 spot check), indicating that the tercile-based findings are not an artifact of the segmentation choice.

The quartile and quintile Δ values are comparable to or slightly larger than tercile values because finer-grained segmentation isolates more extreme phases (the "lowest" quintile captures a more concentrated low-sparsity segment than the lowest tercile). The monotonic dose-response (Δ increasing with s) is preserved across all three segmentation schemes for both models, confirming that the severity gradient is not an artifact of the tercile segmentation choice.

Table 12: Sensitivity of median Δ to `min_phase_length` (mpl). n_{eff}: effective sample size after exclusions. All importance-based $p_{\text{Bonf}} < 10^{-10}$; "ns" denotes $p_{\text{Bonf}} > 0.05$.

Model	Strategy	s	mpl=5	mpl=10	mpl=20	$n_{\text{eff}}(10)$	$n_{\text{eff}}(20)$
DS	Imp	0.15	0.391	0.391	0.393	496	433
DS	Imp	0.20	1.215	1.215	1.235	496	433
DS	Imp	0.25	3.986	3.986	4.017	496	433
DS	Imp	0.30	8.481	8.481	8.420	496	433
DS	Rand	0.15	0.014	0.014^{ns}	0.012^{ns}	496	433
DS	Rand	0.20	0.031^{ns}	0.031^{ns}	0.023^{ns}	496	433
DS	Rand	0.25	0.257	0.257	0.190	496	433
DS	Rand	0.30	0.510	0.510	0.410	496	433
QW	Imp	0.15	0.066	0.066	0.066	475	289
QW	Imp	0.20	0.307	0.307	0.325	475	289
QW	Imp	0.25	1.074	1.074	1.106	475	289
QW	Imp	0.30	3.155	3.155	3.581	475	289
QW	Rand	0.15	-0.011^{ns}	-0.011^{ns}	-0.011^{ns}	475	289
QW	Rand	0.20	0.015^{ns}	0.015^{ns}	0.015^{ns}	475	289
QW	Rand	0.25	0.393	0.393	0.266	475	289
QW	Rand	0.30	0.913	0.913	0.788	475	289

Table 13: Segmentation robustness: median Δ under tercile (n=496/475), quartile (n=50 spot check), and quintile (n=50 spot check) segmentation. Importance-based eviction; all $p_{\text{Bonf}} < 10^{-7}$.

Model	s	Tercile	Quartile [95% CI]	Quintile [95% CI]
DeepSeek	0.15	0.391	0.491 [0.44, 0.70]	0.505 [0.33, 0.71]
	0.20	1.215	1.388 [1.05, 1.79]	1.197 [1.02, 1.76]
	0.25	3.986	4.322 [3.62, 8.06]	4.960 [3.80, 9.02]
	0.30	8.481	12.930 [9.83, 19.10]	11.548 [10.06, 18.51]
Qwen	0.15	0.066	0.226 [0.14, 0.27]	0.203 [0.13, 0.26]
	0.20	0.307	0.561 [0.35, 0.77]	0.517 [0.29, 0.75]
	0.25	1.074	1.870 [1.05, 2.44]	1.801 [1.13, 2.38]
	0.30	3.155	4.097 [2.91, 4.94]	4.009 [3.28, 5.34]